■■ DataSentinel — Pipeline Quality Report

Pipeline: **E-Commerce Orders Validation** | Target Source: orders_csv | Run ID: run-e21f4c84 | Date: 2026-10-30 09:33:41 UTC

Quality Pass Rate	Total Records	Failed Rules	Anomalies	Schema Drift
16.7%	30	4	5	STABLE

1. Quality Rules Validation Breakdown

Status	Rule ID	Type	Column	Failures	Fail %	Details
PASSED	RULE-ORD-001	null_check	order_id	0/30	0.0%	Column 'order_id' has zero null values as required.
FAILED	RULE-ORD-002	unique_check	order_id	2/30	6.7%	Column 'order_id' uniqueness rate 96.67% is below required 100.00%. Found 2 non-...
FAILED	RULE-ORD-003	null_check	customer_email	1/30	3.3%	Column 'customer_email' has 1 null values (3.33%); expected 0.
WARNING	RULE-ORD-004	regex	customer_email	1/30	3.3%	Column 'customer_email' has 1 values that do not match pattern '^[a-zA-Z0-9_.+-]...
FAILED	RULE-ORD-005	range	total_amount	2/30	6.7%	Column 'total_amount' has 2 values outside range (>= 0.01 and <= 10000.0). Sampl...
FAILED	RULE-ORD-006	foreign_key	payment_method	2/30	6.7%	Column 'payment_method' has 2 orphaned/invalid foreign keys not found in set of ...

2. Statistical Anomaly Detection (Z-Score & IQR)

Column	Method	Outliers	Outlier Rate	Distribution Metrics	Top Anomalous Samples
total_amount	ZSCORE	1	3.33%	μ=3392.888, σ=17963.0279	98500.0 (conf=1.0)
total_amount	IQR	2	6.67%	Q1=65.75, Q3=156.25	310.0 (conf=0.5994), 98500.0 (conf=1.0)
item_count	ZSCORE	1	3.33%	μ=2.4667, σ=2.1129	12.0 (conf=1.0)
item_count	IQR	1	3.33%	Q1=1.0, Q3=3.0	12.0 (conf=1.0)